A job ready bootcamp in C++, DSA and IOT

Decision Control instruction



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Agenda

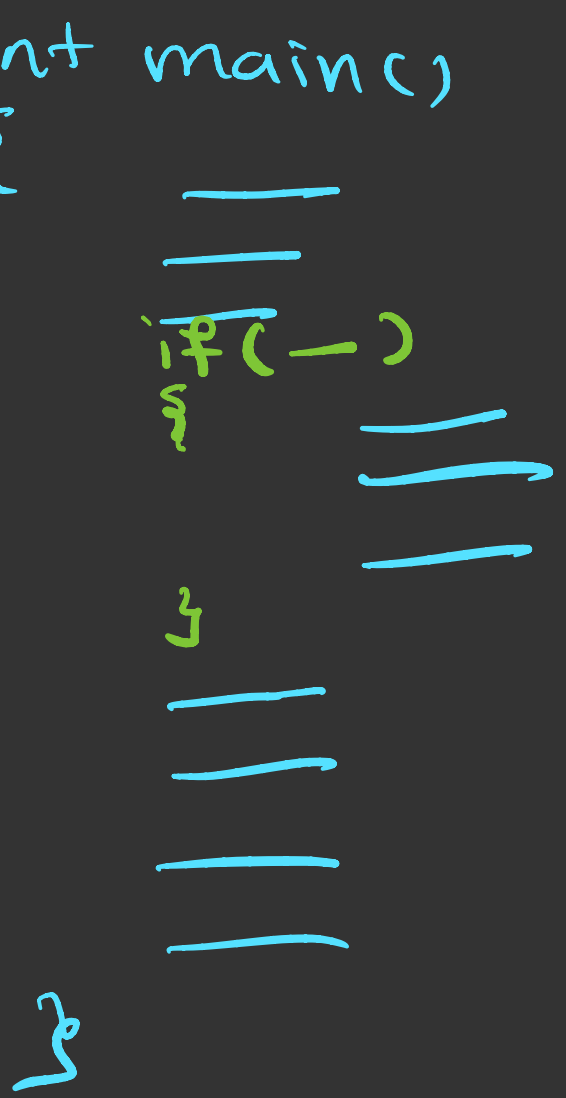
1. Control Statements
2. Decision Control statements
3. if
4. if-else
5. Conditional operator
6. Nested if-else
7. if else ladder

Control Statements

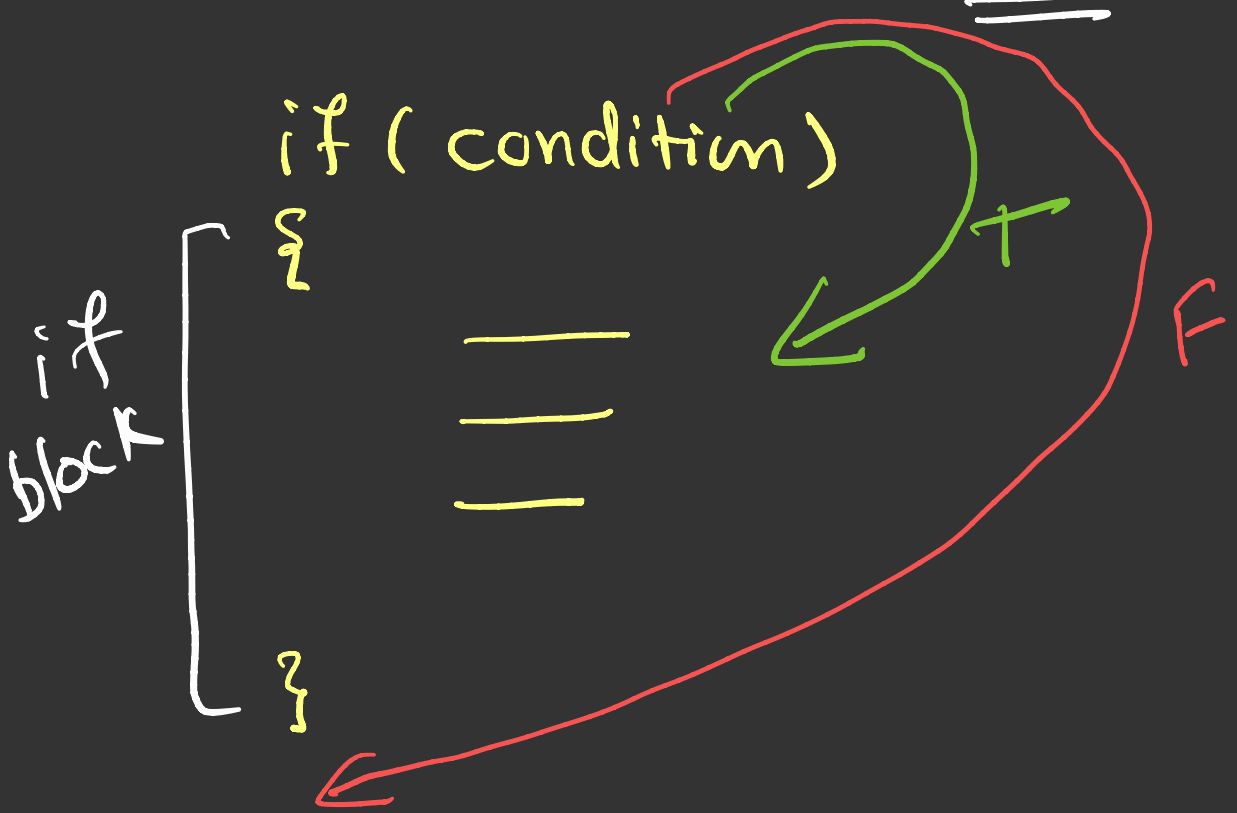
1. Decision Control Statements
2. Iterative Control Statements
3. Switch case Control Statements
4. Goto Control Statements.

Decision Control Statements

- if
- if else
- Conditional Operator



if



यदि or अगर

यदि (TV)

{

पढ़ाई

}

Write a program to check whether a given number is positive or non positive.

5
positive

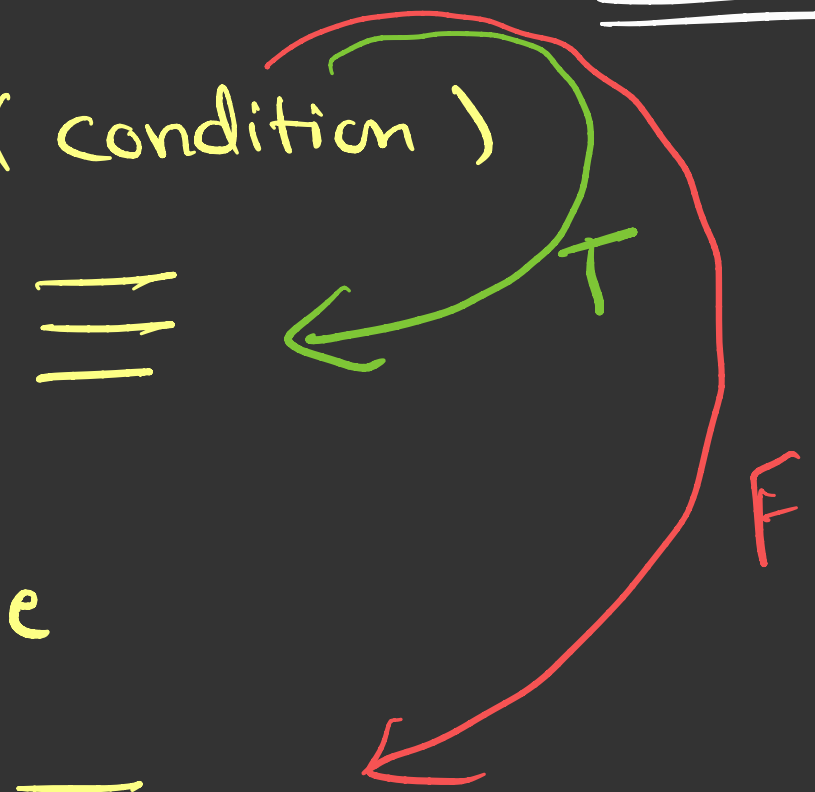
-3
Non-positive

0
Non Positive

1. Understand problem
2. Test case
3. Dry run

if - else

```
if ( condition )
{
    ===
}
else
{
    ===
}
```



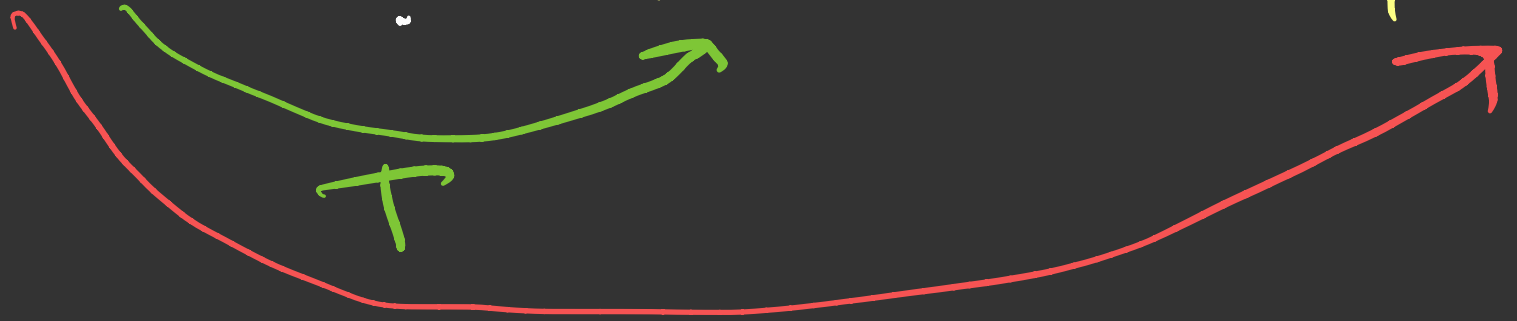

```
यदि (T v)
{
    पढ़ाई
}
नही तो
{
    ज़हर
}
```

Write a program to check whether a given number is a positive or a non positive.

Ternary operator

Conditional Operator

```
expression1 ? expression2 : expression3;
```




```
if (condition)
{
    line1
}
else
{
    line2
}
```

```
condition ? line1 : line2;
```

Write a program to check whether a given number is a positive or a non positive.

What is the difference between conditional operator and if else ?

conditional operator is an expression

and if else is not an expression

```
x = if (-)
    {
      —
    }
    else
    {
      —
    }
```

```
x = — ? — : — ;
```

```
x = a>b ? a : b ;
```

T → x = a

F → x = b

Nested if else

```
if (condition)
{
        if(condition)
        {
            ==
        }
}
```

```
if(condition)
{
        if(condition)
        {
            ==
        }
        else
        {
            ==
        }
}
else
{
        if(condition)
        {
            ==
        }
        else
        {
            ==
        }
}
```

Find output of the following program.

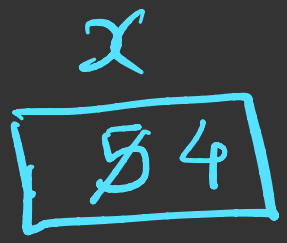
```
int main()
{
        int x = 5;
        if (x > 0)
        {
            printf("Hello");
        }
        printf("%d", x);
}
```

Find Output of the following program.

```
int main()
{
    int x=0;
    if(x)
    {
        printf("Hello");
    }
    printf("%d",x);
}
```

Find Output of the following program.

```
int main()
{
    int x=5;
    if (x=4)
        printf("Hello");
    else
        printf("Bye");
    printf("%d", x);
}
```



(a) Hello5
(b) Bye 5
(c) Hello4
(d) Bye 4

Find output of the following program.

```
int main()
{
    int x=5;
    if (x==4)
        printf("Hello");
        printf("%d",x);
    else
        printf("Bye");
        printf("%d", x);
}
```

if (-)

---

if ( )
{
=
=
3
=
=
else
{
1